# Pedram Beigi, PhD

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## EDUCATION

**The George Washington University** **Sep 2021 – Dec 2026**

*Ph.D., Transportation Engineering (AI, Robotics and AVs)* *GPA: 3.9/4*

*Dissertation: Tactical and Operational Utility-Based Decision-Making Modeling: from Heterogeneous Surface Traffic to Urban Air Mobility, Advisor: Samer Hamdar*

**Sharif University of Technology** **Sep 2016 – Feb 2021**

*B.Sc., Transportation Engineering* *GPA: 3.8/4*

## RESEARCH EXPERIENCE

**University of Illinois Urbana-Champaign** **Urbana, IL**

*Research Intern* *May 2026 – Aug 2026*

- Worked with Prof. Talebpour on developing advanced approaches including foundation models, vision-language architectures, diffusion models, imitation learning, and reinforcement learning, leveraging large-scale driving data.

**The George Washington University** **Washington, DC**

*Graduate Researcher* *Sep 2021 – Dec 2026*

- Developed multi-agent reinforcement learning (MARL) frameworks for AV coordination and negotiation in dynamic partially observable environments using CTDE framework with multi-agent PPO actor-critic models. Applied Bayesian optimization and genetic algorithms for the models' calibration.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments using real-time data (trajectory) streaming using twelve 5G-connected cameras.

**Federal Highway Administration (Turner Fairbank Research Center, Saxton Lab)** **McLean, VA**

*Lead Researcher, Next Generation Modeling for CAVs (Grant ID: 693JJ324C000004)* *Feb 2024 – Present*

- Leading development of trajectory generation models for ADAS (using physics-based, DL-based and RL-based frameworks), and integrating AV behavior models into large-scale microsimulation (SUMO) tools, in collaboration with UIUC and Northwestern University.

*Research Assistant, Automated Vehicles for All (Grant ID: 693JJ319NF00001)* *Sep 2023 – Aug 2024*

- Developed and validated perception (sensor fusion, detection, segmentation) and planning (map generalization, localization) algorithms using LiDAR, radar, and video data for enhanced AV situational awareness and mixed-traffic simulation, in collaboration with Texas A&M, UIUC, UC Davis, NVIDIA and GM.

*Research Assistant, Third Generation Simulation (TGSIM) Data (Grant ID: 693JJ321C000001)* *Sep 2021 – Jul 2023*

- Developed computer vision pipelines to collect large-scale trajectory data (11,000+ vehicle-kilometers) for AV–human interactions, enabling trajectory extraction and motion prediction for AV research.

## TEACHING EXPERIENCE

**The George Washington University**

*Lecturer / TA for CE 1010 (CRN:52745) Intro to Civil Engineering* *Fall 2026*

- Delivered introductory lectures on civil engineering disciplines, transportation systems, and engineering problem-solving; led lab/recitation sessions and graded assignments for first-year engineering students.

*Lecturer / TA for SEAS 1001 (CRN:55178) Engineering Orientation* *Fall 2026*

- Led orientation lectures and discussion sessions introducing SEAS majors, academic pathways, and professional practices; mentored first-year students on course planning and research/career opportunities.

*TA for CN:6730 Sustainable Urban Planning* *Spring 2024, Spring 2025, Fall 2025*

- Supported classroom instruction by developing learning materials, organizing project workflows, and contributing to course content; advised students during office hours and project reviews on research methods.

*TA for CN:4721 Traffic Engineering* *Spring 2025*

- Delivered in-person instructional support, including preparing lecture examples, drafting problem sets, coordinating exams, and helping shape course modules related to traffic operations and roadway analysis.

## RESEARCH INTERESTS

My research interests focus on traffic flow theory, behavior modeling, CAVs, simulation, machine learning, and computer vision to enhance safety, efficiency, and resilience in transportation systems:

- **Machine Learning and Artificial Intelligence:** Developing deep reinforcement learning architectures, including multi-agent reinforcement learning, foundation models, vision-language models, diffusion models, and imitation learning for autonomous driving applications.
- **Automated Driving Systems (ADS) and Trajectory Planning:** Designing perception, prediction, and planning pipelines for autonomous vehicles using multi-sensor fusion. Focus areas include generating trajectory models for ADAS and enhancing situational awareness in mixed-traffic environments.
- **Traffic Flow Theory and Traffic Simulation:** Macroscopic and microscopic traffic flow analysis, integrating data-driven and physics-based behavior models into large-scale microsimulation tools. Building integrated simulation pipelines for closed-loop decision-making using real-time data streaming.
- **Computer Vision and Large-Scale Data Modeling:** Engineering computer vision pipelines to extract large-scale trajectory data for AV-human interaction modeling and motion prediction. Utilizing multi-modal deep learning time-series models to monitor driver behavior and deploy real-time IoT alerts.
- **Transportation Operations and Safety Optimization:** Applying Bayesian optimization, genetic algorithms, and reinforcement learning to calibrate traffic models and evaluate vulnerable road user behavior. Analyzing the impact of V2X connectivity on intersection performance and overall network reliability.

## PROPOSAL WRITING EXPERIENCE

### U.S. Department of Transportation

*FHWA Exploratory Advanced Research (EAR) Program, Solicitation No. 693JJ3-24-BAA-0001*

- Title: New Generation Models for Connected Autonomous Vehicles: NGM-CAV
- Proposer: The George Washington University (Lead), Northwestern University (Subcontractor), University of Illinois Urbana-Champaign (Subcontractor)
- Status: Awarded
- Budget: $432,717 for the project and $344,881 allocated to The George Washington University
- Announcement and Synopsis: https://www.highergov.com/contract/693JJ324C000004/

## INDUSTRY EXPERIENCE

**Mead&Hunt** **May - Aug 2025**

*Advanced Mobility Intern* *Washington, DC*

- Assisted the lead engineer in citywide traffic signal optimization and timing plan implementation for DDOT and BCDOT, utilizing Synchro, and signal controller data to enhance corridor coordination and reduce travel times.

- Supported development of traffic simulation and forecasting models for smart mobility and corridor planning projects with WVDOT.

**moment.ai** **Sep - Dec 2024**

*Autonomous Vehicle Researcher Intern* *Washington, DC*

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

**Arcadis** **May - Aug 2024**

*Transportation Engineer Intern* *Atlanta, GA*

- Collaborated with GDOT on concept design, traffic volume and capacity analysis, operational analysis, and the development of a mitigation plan for the I-285 Express Lanes in Atlanta.
- Conducted meetings with stakeholders, writing reports, and assisting in the preparation of presentations for project reviews and approvals.

## HONORS AND AWARDS

COMTO DC Scholarship 2026
ITE CAPSITE Pat Timbrook Scholarship 2025
ITS Maryland Student Scholarship 2025
FGSS and SEAS Tuition Award, The George Washington University

## TECHNICAL SKILLS

**Programming:** Python, ROS, MySQL, Git
**Libraries:** PyTorch, Diffusers, Transformers,OpenCV, RLlib, Gymnasium, Scikit-learn, Keras, Ultralytics
**Tools:** CARLA, Unity, Power BI, MicroStation, Tableau, SUMO, VISSIM, Synchro, SIDRA

## PUBLICATIONS (h-index: 6; Citations: 271)

T. Petrov, W. Sleiman, P. **Beigi**, P. Počta, L. Buzna, and S. Hamdar, "Simulation of vehicle-to-pedestrian communication for development of novel c-its services for vulnerable road users," *Transportation Research Procedia*, vol. 93, pp. 674–679, 2026 [DOI]

W. Sleiman, M. Haque, M. S. Amin, P. **Beigi**, M. Khoueiry, S. Kang, and S. Hamdar, "Diffusion process-based model for network trajectory propagation: Formulation, data mining, and cross-comparative analysis," *IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)*, 2026 [DOI]

W. Sleiman, P. **Beigi**, and S. Hamdar, "Percolation-Based Cloud Computing Framework for Network Reliability Assessment," in *IEEE 4th Int. Conf. on Computing and Machine Intelligence (IEEE ICMI)*, pp. 1–7, IEEE, 2025 [DOI]

P. **Beigi**, S. Hamdar, A. Talebpour, and H. Mahmassani, "Impact of Heavy Vehicles on Headway Distributions: Study Using Naturalistic Urban Expressway Trajectories," *Transportation Research Record (TRR)*, pp. 1–15, 2025 [DOI] [Code]

D. Pan, P. **Beigi**, W. Daamen, V. Knoop, A. Talebpour, and S. Hamdar, "Exploring Voronoi-based area-weighted network fundamental diagrams in shared urban spaces," *Transportmetrica A: Transport Science*, pp. 1–27, 2025 [DOI]

W. Sleiman, P. **Beigi**, T. Petrov, L. Buzna, P. Pocta, and S. Hamdar, "Impact of Pedestrian and Vehicle Connectivity on Intersection Performance: A High-Fidelity Simulation with Deep Reinforcement Learning Control," *Transportation Research Record (TRR)*, pp. 1–15, 2025 [DOI] [Code]

R. Ammourah, P. **Beigi**, B. Fan, S. H. Hamdar, *et al.*, "Introduction to the Third Generation Simulation Dataset: Data Collection and Trajectory Extraction," *Transportation Research Record (TRR)*, pp. 241–257, 2024 [DOI] [Code] [Data]

F. M. Rad, M. Beygi, P. **Beigi**, and A. Samimi, "Predicting Freight Attraction with MLR and GWR using VIIRS Satellite Nighttime Imagery Data," *Advances in Transportation Studies*, vol. 64, 2024 [DOI]

F. Faghihinejad, M. Fard, A. Roshanghalb, and P. **Beigi**, "A framework to assess the correlation between transportation infrastructure access and economics," *Mathematical Problems in Engineering*, vol. 2022, no. 1, 2022 [DOI]

P. **Beigi**, M. Haque, M. S. Rajabi, and S. Hamdar, "Bike share's impact on covid-19 transmission and bike share's responses to covid-19," in *Canadian Society of Civil Engineering Annual Conference (CSCE)*, pp. 723–737, Springer, 2022 [DOI]

P. **Beigi**, M. Khoueiry, M. S. Rajabi, and S. Hamdar, "Station reallocation and rebalancing strategy for bike-sharing systems," in *Canadian Society of Civil Engineering Annual Conference (CSCE)*, pp. 621–630, Springer, 2022 [DOI]

## BOOK CHAPTERS

P. **Beigi**, M. S. Rajabi, and S. Aghakhani, "An Overview of Drone Energy Consumption Factors and Models," *Handbook of smart energy systems*, pp. 1–20, 2022 Springer Nature© [DOI]

M. S. Rajabi, P. **Beigi**, and S. Aghakhani, "Drone Delivery Systems and Energy Management: A Review and Future Trends," *Handbook of smart energy systems*, pp. 1–19, 2023 Springer Nature© [DOI]

## UNDER REVIEW PUBLICATIONS

P. **Beigi**, W. Sleiman, A. Khakpour, M. Khoueiry, and S. Hamdar, "Motion as Communication: Plan-Conditioned Diffusion World Models for Negotiation at Unprotected Conflict Points." Manuscript under review at IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS), 2026 [Code]

P. **Beigi**, A. Talebpour, and S. H. Hamdar, "Utility-Guided Residual Multi-Agent Reinforcement Learning for Closed-Loop Interactive Driving." Manuscript submitted to Transportation Research Part C: Emerging Technologies (TR-C); under review, 2026 [Code]

M. S. Khoueiry, P. **Beigi**, and S. H. Hamdar, "Toward Autonomous Driving in Weak-Lane Freeway Traffic: A Naturalistic Trajectory Dataset and Comparative Evaluation of Interaction Models." Manuscript under review at Transportation Research Part C: Emerging Technologies (TR-C), 2026 [Code]

A. Khakpour, N. Li, P. **Beigi**, and Y. M. Nie, "Empirically Calibrated Differential Games for Simulating Driver Interactions," 2026. Manuscript under review at IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS) [arXiv] [Data]

S. J. Hegde, A. Khakpour, P. **Beigi**, Y. M. Nie, and H. S. Mahmassani, "Complete Street Design informed by Multi-Agent Simulation," 2026. Under review at Transportation Research Record (TRR) [arXiv]

S. Bafandkar, P. **Beigi**, N. Li, J. Houdros, S. H. Hamdar, H. S. Mahmassani, and A. Talebpour, "Learning From Real-World Trajectories: A Hybrid Genetic Algorithm And Reinforcement Learning Approach For Vulnerable Road User Model Calibration," *Transportation Research Board - Under Review (TRB)*, 2026 [Code]

W. Sleiman, P. **Beigi**, T. Petrov, L. Buzna, P. Pocta, and S. Hamdar, "Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Signal Control," in *IEEE Vehicular Technology Conference (IEEE VTC) - Under Review*, 2026 [Code]

## CONFERENCE PRESENTATIONS (*presenting author)

P. ***Beigi***, N. Li, S. Bafandkar, J. Hourdos, A. Talebpour, and S. H. Hamdar, "NGM: An Open, Modular, and Extensible Framework for Trajectory-Calibrated Microsimulation of Mixed Human, Automated, and Connected Traffic." Submitted to the 2027 Transportation Research Board (TRB) Annual Meeting, Presentation and Publication track; under review; manuscript TRBAM-27-06429, 2026 [Code]

P. ***Beigi***, N. Li, J. Hourdos, A. Talebpour, and S. H. Hamdar, "Operations and Safety Impacts of Connected and Automated Vehicles under Imperfect V2V, Mixed Fleets, and Intersection Conflict Exposure." Submitted to the 2027 Transportation Research Board (TRB) Annual Meeting, Presentation track; under review; manuscript TRBAM-27-06079, 2026

S. J. *Hegde, A. Khakpour, P. **Beigi**, Y. M. Nie, and H. S. Mahmassani, "Complete Street Design informed by Multi-Agent Simulation." Submitted to the 2027 Transportation Research Board (TRB) Annual Meeting, Presentation and Publication track; under review; manuscript TRBAM-27-04884, 2026

A. *Khakpour, N. Li, P. **Beigi**, H. Kobeissi, and Y. M. Nie, "A Game-Theoretic Approach to Modeling Human-Driven Vehicle Interactions at Urban Intersections." Submitted to the 2027 Transportation Research Board (TRB) Annual Meeting, Presentation track; under review; manuscript TRBAM-27-04886, 2026

M. E. *Rashidi, W. Sleiman, P. **Beigi**, M. S. Khoueiry, W. Zhang, and S. H. Hamdar, "An Interaction-Aware Kinematic-Historic Transition Model for Weak-Lane Microscopic Traffic Simulation." Submitted to the 2027 Transportation Research Board (TRB) Annual Meeting, Presentation track; under review; manuscript TRBAM-27-03861, 2026

P. ***Beigi***, W. Sleiman, A. Khakpour, M. Khoueiry, and S. Hamdar, "Motion as Communication: Plan-Conditioned Diffusion World Models for Negotiation at Unprotected Conflict Points," in *IROS 2026 Workshop on Bridging the Gap between Neural and Symbolic World Models for Robot Planning, Reasoning, and Action (RoBoWoMo)*, 2026. [Paper] [Code]

M. S. Khoueiry, P. ***Beigi***, A. Talebpour, and S. H. Hamdar, "Bridging Perception, Planning and Control: A System-Level Validation of a Microscopic Acceleration Model for Longitudinal Autonomous Driving," in *IROS 2026 Workshop on Leveraging Behavioral Models in Human-Autonomy Teams (BEMHAT)*, 2026. [Paper] [Code]

A. *Khakpour, P. **Beigi**, N. Li, S. Hegde, and H. Mahmassani, "Understanding the Interaction of Vehicles within an Intersection: An Optimal Control Approach with Differential Games," in *INFORMS*, (San Francisco, CA, USA), November 2026

P. ***Beigi***, N. Li, S. Bafandkar, D. Monzer, J. Hourdos, H. Mahmassani, A. Talebpour, and S. Hamdar, "A data-driven comparison of car-following behaviors of autonomous and human-driven vehicles and their traffic flow impacts under naturalistic traffic conditions," in *Transportation Research Board Annual Meeting (TRBAM-26-04796) (TRB)*, (Washington, D.C., USA), January 2026 [DOI]

S. *Bafandkar, P. **Beigi**, N. Li, J. Houdros, S. H. Hamdar, H. S. Mahmassani, and A. Talebpour, "Learning from real-world trajectories: A hybrid genetic algorithm and reinforcement learning approach for vulnerable road user model calibration," in *Transportation Research Board Annual Meeting (TRBAM-26-04981) (TRB)*, (Washington, D.C., USA), January 2026 [DOI]

D. *Pan, M. Haque, A. P. Nujjetty, P. **Beigi**, W. Zhang, W. Sleiman, and S. Hamdar, "Calibrating traffic models: Analyzing vehicles and vulnerable road users interactions at roundabouts using real-world trajectories," in *Transportation Research Board Annual Meeting (TRBAM-26-04076) (TRB)*, (Washington, D.C., USA), January 2026

W. *Sleiman, T. Petrov, P. **Beigi**, L. Buzna, P. Pocta, and S. Hamdar, "Adaptive traffic signal control under partial connectivity co-simulation and penetration-aware policy learning," in *Transportation Research Board Annual Meeting (TRBAM-26-02575) (TRB)*, (Washington, D.C., USA), January 2026

W. *Sleiman, P. **Beigi**, N. Li, and Y. Zhang, "A lifelong devotion to transportation research: Hani's contribution in numbers," in *Memorial and Tribute for Professor Hani S. Mahmassani in Transportation Research Board Annual Meeting (TRB)*, (Washington, D.C., USA), January 2026 [Booklet]

W. *Sleiman, P. **Beigi**, and S. Hamdar, "Percolation-Based Cloud Computing Framework for Network Reliability Assessment," in *IEEE 4th Int. Conf. on Computing and Machine Intelligence (IEEE ICMI)*, (Mount Pleasant, MI, USA), April 2025

W. *Sleiman, P. **Beigi**, T. Petrov, L. Buzna, P. Pocta, and S. Hamdar, "High fidelity simulation framework for deep reinforcement learning control: Impact of connected pedestrian and vehicle penetration rates on intersection signalization performance," in *Transportation Research Board Annual Meeting (TRBAM-25-04809) (TRB)*, (Washington, D.C., USA), January 2025 [TRID] [Code]

D. *Pan, P. **Beigi**, S. Hamdar, W. Zhang, and W. Sleiman, "Leveraging naturalistic trajectory data for safety analysis of all-way stop control intersections: Exploration speed limit and stop violations patterns," in *Transportation Research Board Annual Meeting (TRBAM-25-05015) (TRB)*, (Washington, D.C., USA), January 2025

M. *Khoueiry, M. Haque, P. **Beigi**, A. Talebpour, and S. Hamdar, "Non-lane-based acceleration behavior on a freeway in jounieh, lebanon: From aerial trajectory extraction to model calibration," in *Transportation Research Board Annual Meeting (TRBAM-25-05607) (TRB)*, (Washington, D.C., USA), January 2025

P. ***Beigi**, M. Haque, W. Sleiman, J. Hourdos, H. Mahmassani, A. Talebpour, and S. Hamdar, "Calibrating car-following behaviors for automated and human-driven vehicles under naturalistic traffic conditions: Intelligent driver model parametric differences," in *Transportation Research Board Annual Meeting (TRBAM-25-06282) (TRB)*, (Washington, D.C., USA), January 2025

P. ***Beigi**, S. H. Hamdar, R. Ammourah, A. Talebpour, and H. Mahmassani, "The impact of heavy vehicles on headway distributions: A study using naturalistic urban expressway trajectories," in *Transportation Research Board Annual Meeting (TRBAM-24-04398) (TRB)*, (Washington, D.C., USA), January 2024 [TRID] [Code]

D. *Pan, P. **Beigi**, S. Hamdar, W. Sleiman, and W. Zhang, "Calibrating traffic models: Analyzing vehicles and vulnerable road users interactions at roundabouts using real-world trajectories," in *TRB 7th International Conference on Roundabouts and Geometric Design (Lectern Session: Designing for Pedestrians and Bicyclists)*, (Atlanta, GA, USA), June 2024

P. ***Beigi**, D. Pan, S. Hamdar, and W. Zhang, "Big data analysis for los extraction through naturalistic trajectories: A shrp2 study for signalized intersections," in *2024 Joint ITE International & Mid-Colonial District Annual Meeting (Session: Traffic Analysis Approaches for Intersections and Networks)*, (Philadelphia, PA, USA), July 2024

R. Ammourah, P. **Beigi**, B. Fan, S. H. Hamdar, R. James, M. Khajeh-Hosseini, H. Mahmassani, D. Monzer, and A. *Talebpour, "Introduction to the third generation simulation (tgsim) dataset: Data collection and trajectory extraction," in *Transportation Research Board Annual Meeting (TRBAM-23-03535) (TRB)*, (Washington, D.C., USA), January 2023 [TRID]

B. *Fan, R. Ammourah, P. **Beigi**, S. H. Hamdar, H. Mahmassani, and A. Talebpour, "Formulation and calibration of a hybrid car-following model for automatic cruise controlled cars in actual highway traffic," in *Transportation Research Board Annual Meeting (TRBAM-23-04567) (TRB)*, (Washington, D.C., USA), January 2023

P. ***Beigi**, S. H. Hamdar, B. Fan, R. Ammourah, A. Talebpour, and H. S. Mahmassani, "From videography to traffic analysis: Exploring the heavy vehicle impact on headway distributions using extracted high resolution trajectory data on a washington dc expressway," in *Transportation Research Board Annual Meeting (TRBAM-23-04343) (TRB)*, (Washington, D.C., USA), January 2023

## Professional Affiliations and Review Activities

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|---|---|
| Editorial Board Member, *Journal of Future Sustainability* | Canada |
| Reviewer, *Conference on Neural Information Processing Systems (NeurIPS)* | International |
| Reviewer, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* | International |
| Reviewer, *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* | United States |
| Reviewer, *Transportation Research Part C: Emerging technologies* | United Kingdom |
| Reviewer, *IEEE Transactions on Intelligent Transportation Systems Journal* | United States |
| Reviewer, *Expert Systems With Applications* | United Kingdom |
| Reviewer, *Transportmetrica B: Transport Dynamics* | United Kingdom |
| Reviewer, *Transportation Planning and Technology Journal* | United Kingdom |
| Reviewer, *Traffic Injury Prevention* | United Kingdom |
| Reviewer, *Transportation Research Board (TRB)* | United States |
| Reviewer, *Transportation Research Record (TRR)* | United States |
| Reviewer, *American Journal of Traffic and Transportation Engineering* | United States |
| Reviewer, *Journal of Intelligent Transportation Systems* | United Kingdom |
| Reviewer, *Canadian Journal of Civil Engineering* | Canada |
| Reviewer, *Advances in Transportation Studies Journal* | Italy |
| Reviewer, *Transportation Safety and Environment* | United Kingdom |
| Reviewer, *Modelling and Simulation in Engineering* | United Kingdom |
| Reviewer, *Sustainable Cities and Society* | Netherlands |

## Voluntary Affiliations

Member, Google Developer Student Club at GW
President, Institute of Transportation Engineers (ITE) Student Chapter at GW
Member, Institute of Electrical and Electronics Engineers (IEEE)
Member, Intelligent Transportation Society of Maryland (ITSMD)
Member, Conference of Minority Transportation Officials (COMTO)

## REFERENCES

- **Samer Hamdar, PhD**
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  *The George Washington University*
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- **Alireza Talebpour, PhD**
  *Associate Professor, Department of Civil and Environmental Engineering*
  *University of Illinois Urbana–Champaign*
  *Email: ataleb@illinois.edu — Phone: 847-644-2870*

- **Yu (Marco) Nie, PhD**
  *Professor and Roxelyn and Richard Pepper Family Chair*
  *Department of Civil and Environmental Engineering*
  *Northwestern University*
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  *Assistant Professor, Department of Civil and Environmental Engineering*
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  *Professor, Charles E. Via Jr. Department of Civil and Environmental Engineering*
  *Courtesy Professor, Bradley Department of Electrical and Computer Engineering*
  *Director, Center for Sustainable Mobility*
  *Virginia Polytechnic Institute and State University*
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